YAO-CHING HUANG

M.S. in Electronic Engineering, NTUST | Mobile Information Technology Lab (MITLab)
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Programming Languages	Python, SQL, Cython, JavaScript
Full-Stack & Database	React, Django, REST API, PostgreSQL, MariaDB, MySQL
AI & Agent	Dify, Multi-Agent, ReAct, LangGraph, MCP, CLI, RAG, n8n
Cloud Infrastructure	Linux, Docker, Kubernetes, Nginx, Harbor, Rancher, Harvester
DevOps	Git, GitHub Actions, CI/CD
GPU Computing	NVIDIA MPS, CUDA, NVIDIA DCGM
Monitoring & Testing	Prometheus, Istio, eBPF
Digital Twin & Simulation	NVIDIA Omniverse, NVIDIA Sionna RT
5G & Networking	O-RAN, 5G NR, OpenAirInterface (OAI), E2AP

Education

NTUST | M.S. in Electronic Engineering | GPA 4.18 / 4.30 — Sep 2025 – Jul 2027 (expected)

NTUST | B.S. in Electronic Engineering | GPA 4.04 / 4.30 — Sep 2021 – Jun 2025

Class rank **3 / 29** | Department rank **10 / 104**

Research & Industry Projects

NSTC Four-Year 6G Initiative | AI/ML Platform & AI Agents — Jun 2025 – Present

- **Platform Integration:** Integrated **20+ Docker/Kubernetes** services and **Kubeflow/Argo** workflows to support AI/ML operations.
- **Platform Reliability:** Built monitoring with **Prometheus/NVIDIA DCGM**, developed nine automated stress-test suites, and resolved **seven** platform-level issues.
- Designed an **NVIDIA MPS** GPU scheduler, increasing aggregate throughput by **21–24%** and reducing batch completion time by **13%**.
- Developed **Dify Multi-Agent ReAct** workflows with intent-aware tools and backend execution logic for automated model training, monitoring, and diagnosis.

ITRI | 5G NR RAN Digital Twin & O-RAN Integration — Apr 2026 – Present

- Developed an **E2 Adapter** connecting simulated RAN to **Near-RT RIC**; integrated **NVIDIA Omniverse/Sionna RT** for 3D radio-channel simulation.
- Achieved **90–96%** agreement with **OAI rfsim** metrics; reduced mean simulation error from **50.1%** to **2.5%** using ML calibration.

Professional Experience

SparkAmplify (California, USA) | Backend / DevOps Engineering Intern — Jan – Mar 2026

- Developed **REST APIs**, database **schemas**, and **LLM API** chatbot integration for the MySphere platform.
- Deployed containerized services using **Docker/Nginx/Harbor** and automated image builds and deployment with **GitHub Actions CI/CD**.

NTUST | Systems Operations & Database Administrator — Sep 2026 – Present

- Maintained **PHP/MariaDB** systems, developed and extended backend features, and resolved system and user-reported issues.

NTUST | Calculus Teaching Assistant (EMI) — Sep 2025 – Present

- Built a course system with **React/Django** and a **RAG-powered AI Agent** for automated exam and Word document generation.

Ying Na Industrial Co., Ltd. | IT Systems Administrator — Aug 2025 – Present

- Maintained servers, databases, internal IT systems and backups; handled equipment procurement, licensing, and employee IT support.

Awards & International Activities

- **2026 Hanyang University 6G AI-RAN/O-RAN Workshop:** Deployed **Open5GS, srsRAN, FlexRIC** and a KPM monitoring xApp.
- **2026 Intelligent Networking SIG Seminar:** Presented **6G RAN Digital Twin** and **AI Agent** demonstrations.
- **2025–2026 MOEA Space & Networking R&D Program:** Selected in two consecutive years; received a **2025 Outstanding Performance** certificate.

Selected Projects

1st NTUST Youth Startup Hackathon — Aug 2026

- Entry **FairWinds** — an IoT fan-sharing rental service. Built the **ESP32** locker firmware and cloud integration, **backend APIs and database schema**, and frontend animation wiring; a full rent-to-return prototype in three days.

LLM-Powered Productivity Assistant

- Built **LangGraph/ReAct** agent workflows with short- and long-term memory, **MySQL** queries, and **MCP** tools for email and calendar tasks.

SSD Garbage Collection Scheduling Optimization

- Studied **Deep Q-Learning** scheduling; connected Python to an existing C++ simulator via **Cython** and evaluated 100,000 I/O traces.